ADITYA CHUNDURI

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chunduri-aditya.github.io/Portfolio

EDUCATION

University of Southern California — Los Angeles, CA
Master of Science in Applied Data Science — *January 2024 – December 2025*

SRM Institute of Science and Technology — Chennai, India
Bachelor of Technology in Computer Science and Engineering (AI/ML Specialization) — *June 2019 – July 2023*

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript, Bash
ML and AI Frameworks: PyTorch, TensorFlow, Scikit-learn, Hugging Face, Keras, OpenCV, Librosa, Demucs, LangChain
LLM Evaluation and Safety: Inspect AI, AgentDojo, Ollama, ChromaDB, RAG, DPO, adversarial evaluation, prompt injection
MLOps and Cloud: Docker, AWS, FastAPI, Flask, Git, GitHub Actions, CI/CD, Gradio
Data and Tools: PostgreSQL, Pandas, NumPy, Plotly, ChromaDB

EXPERIENCE

USC – Viterbi School of Engineering — Los Angeles, CA
Research Assistant, Computer Vision and Medical Imaging — *August 2024 – December 2024*

- Engineered a U-Net segmentation pipeline in PyTorch for retinal artery-vein classification on the CVD-Masks dataset, achieving 94% Dice score through architectural tuning and custom loss functions.
- Designed modular PyTorch training pipelines with experiment tracking across 3 concurrent projects, reducing iteration cycles by 30% and enabling reproducible model comparisons.
- Built an automated medical image pipeline with quality-validation controls, cutting manual curation time by 40% across 10,000+ images.

SSN College of Engineering — Remote
Research Intern, Computer Vision and Object Detection — *June 2021 – July 2021*

- Built real-time YOLO-based object tracking pipelines at 30+ fps (15% detection robustness gain across 5 conditions); shipped 10+ reusable PyTorch/TensorFlow modules eliminating manual frame-annotation overhead.

PROJECTS

Agent Shield: LLM Agent Security Evaluation Framework | *Python, Inspect AI, AgentDojo* — *April 2026 – Present*

- Designing an 8-module adversarial evaluation framework on the Inspect AI harness (UK AISI) for prompt injection, MCP tool poisoning, RAG memory attacks, multi-agent manipulation, and Cialdini-grounded behavioral drift.
- Benchmarking 8 frontier models (Claude, GPT-4.1, Gemini 2.5, Llama 3.1, Mistral Large, DeepSeek V3) with unified attack-success-rate (ASR) and utility-under-attack metrics; arxiv preprint in progress.

AI RemixMate: AI-Powered DJ Mixing Engine | *Python, PyTorch, Librosa, Demucs, Gradio* — *September – December 2025*

- Engineered an end-to-end audio ML pipeline: Demucs separation, Librosa spectral features, Krumhansl-Schmuckler key detection, Camelot Wheel matching, and FAISS vector indexing; deployed via FastAPI with GPU acceleration.
- Shipped a Gradio web UI supporting 8+ audio formats with real-time BPM, key, and energy analysis; open-sourced with 9 GitHub stars.

AkashicTree: Automated Content Generation Pipeline | *Python, Diffusers, Ollama, ElevenLabs, FLUX.1* — *June – August 2025*

- Designed a model-agnostic multimodal generation pipeline orchestrating text (Ollama), image (FLUX.1/Diffusers), and audio (ElevenLabs) models from a single input brief, cutting production time by 80%.
- Implemented tiered backend switching between local Ollama inference and cloud APIs (GPT-4o-mini), reducing per-iteration inference cost by 90% while maintaining output quality.

AI Health Journal: Privacy-First LLM Assistant | *Python, Flask, Ollama, ChromaDB* — *January – May 2025*

- Independently built a production RAG system: ChromaDB vector store, Ollama 7B-parameter local inference, Flask API layer; zero external API dependencies.
- Applied DPO fine-tuning for domain-specific health response alignment; optimized vector retrieval achieving sub-second context injection across 10,000+ document chunks.

Model-Behavior-Lab: Adversarial LLM Evaluation Platform | *Python, Ollama, Plotly, CI/CD* — *September – December 2024*

- Built an adversarial LLM evaluation platform running 5,000+ tests across 10+ frontier models (GPT-4, Claude, Llama, Mistral) measuring instruction-hierarchy violations, hallucination rate, and refusal consistency; CI/CD-integrated JSON test suites cut evaluation setup from hours to under 5 minutes.

PUBLICATION **Wind Power Analysis Using Machine Learning** — IJRASET, Aug 2023 (Co-Author).